

IA – 2

Branch	EXCP,CSBS,VLSI	Semester	IV
Course Name & Course Code	Foundations of Generative AI 216H28C401	Academic year	JAN 2026 – MAY 2026
Faculty In-charges	Dr. Tejashree Suresh Phatak	Year	SY B.Tech

Sr No	All questions are compulsory. Each question carries 5 marks.	Marks
Q1	Explain N gram Modelling	5
Q2	Explain Text Processing steps in Natural Language Processing	5
Q3	State any two real-world application of Generative AI	5
Q4	Suppose we have a small corpus: "I love AI" "I love machine learning" "I love deep learning" Calculate Probability : $p(\text{machine/love})$	5

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Ans 1:] N-gram modelling is a probabilistic languages model used in natural language processing to predict the next word in a sequence based on the previous $n-1$ words. An N-gram is a sequence of 'n' words, such as unigram, bigram and trigram. It works on the Markov assumptions which states that the probability of a word depends only on a limited number of preceding words rather than the entire sentence. For example, in a bigram model, the probability of "learning" depends only on the word "machine". N-gram models are simple yet effective and widely used in applications like text prediction, speech recognition, and machine translation.

Ans 2:] Text processing in NLP involves transforming raw text into a structured format the machines can analyze. This process starts with tokenization, where text is divided into words or sentences. Then the text is converted to lowercase to maintain uniformity. Stopwords such as "the" and "a" are removed to reduce unnecessary data. Words are then reduced to their base form using stemming or lemmatization for example, "running" becomes "run". Additional steps like part-of-speech tagging help like identify grammatical roles of words. Finally, the text is converted into numerical representation such as TF-IDF or vectors so that machine learning models can process it effectively.

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Ans3.] Generative AI has many real-world applications in different industries. One major application is content generation, where AI systems can generate text such as articles, reports, code and emails. This helps automate tasks and improve productivity. Another important application is image generation, where AI can create realistic images from text descriptions, which is useful in design, advertising and entertainment. These applications show how generative AI enhances creativity & efficiency by producing high-quality content automatically.

Ans4.] Given the corpus "I love AI", "I love machine learning" and "I love deep learning" we use the bigram model to calculate probability. The probability $P(\text{machine} | \text{love})$ is calculated as the count of the phrase "love machine" divided by the total count of the word "love". In the corpus "love machine" appears once, while "love" appears three times. Therefore, the probability is $1/3$, which is approximately 0.33. This shows the likelihood of the word "machine" occurring after "love" in the given data.

Corpus: "I love AI".

"I love machine learning".

"I love deep learning".

We use Bigram Probability:

$$P(\text{machine} | \text{love}) = \frac{\text{count}(\text{love, machine})}{\text{count}(\text{love})}$$

$$\text{Step 1: "love machine"} = 1 \quad \text{count}(\text{love})$$

$$\text{"love" appears} = 3$$

Step 2: Calculate probability

$$P(\text{machine} | \text{love}) = \frac{1}{3}$$

$$\therefore P(\text{machine} | \text{love}) = 0.33$$

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